Department of Physics
Faculty of Science
University of Yaoundé I
Yaoundé, Cameroon

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The Editorial Board
Chemical Science
Royal Society of Chemistry

Dear Editors,

We are pleased to submit our manuscript, "Quantum-Coherent Control of Biological Energy Transfer via Spectral Bath Engineering," for consideration as an Article in Chemical Science.

What we found: The longstanding assumption in quantum biology is that the noisy protein-solvent environment of light-harvesting systems must be accepted as an evolutionary fixed boundary condition. By deploying the numerically exact Process Tensor Hierarchy of Pure States (PT-HOPS), we demonstrate that *external* spectral shaping of the incident photon bath represents a novel mechanism for active biophysical control. Our simulations confirm that subjecting the Fenna–Matthews–Olson (FMO) complex to an engineered dual-band (750 nm and 820 nm) optical filter selectively amputates destructive pure dephasing modes via polaron-dressing. This successfully extends excitonic coherence lifetimes by 20 % to 50 % and boosts reaction center energy transfer efficiency by 25 % under physiological conditions.

Why it matters: Our findings physically shift the paradigm of Environment-Assisted Quantum Transport (ENAQT) from a passive survival trait to an actively controllable variable. The demonstrated 50 % temporal extension of quantum superpositions *in vivo* proves that coherent transport states can be manipulated externally without chemical mutation. These results carry immense implications for designing artificial, biomimetic light-harvesting systems, structurally protected quantum logic bio-implants, and optimized artificial photosynthesis cascades.

Why Chemical Science: This manuscript bridges the critical interfaces between biophysical chemistry, non-Markovian quantum dynamics, and advanced materials engineering. We firmly believe Chemical Science's multidisciplinary, high-impact readership is uniquely poised to leverage our findings detailing how fundamental light-matter manipulation governs complex molecular aggregates. The 12-test algorithm validation benchmarking ensures the chemical physics foundations are robust, while the biological consequences address profound systemic challenges.

Statement of Originality: All authors have explicitly approved this manuscript. We declare no competing financial disputes or conflicts of interest. This manuscript has not been submitted or published elsewhere.

Thank you for your consideration.

Sincerely,

Steve Cabrel Teguia Kouam

Steve Cabrel Teguia Kouam Steve Cabrel Teguia Kouam
Department of Physics
University of Douala, Cameroon
Email: steve.teguia@facsciences-uy1.cm